AUTONOMOUS ML RESEARCH · TECHJAM 2026

SciOdyssey: 3D Spatial Dimension

Long-Horizon Autonomous ML Discovery through **Pure Evidence** & **Decoupled Lifetimes**.

PERSISTENT WORLD

FRESH SCIENTIST

EVIDENCE FIRST

01 / PROBLEM · THE RESEARCH TASK

Can an Agent Carry a Research Task to Completion?

Move beyond one-off script generation: turn an open-ended research contract into a validated model with verifiable evidence.

CAPABILITY 01

Autonomous Exploration

Formulate original hypotheses, write model code, design feature interactions, and execute GPU training runs without handholding.

Scientific Judgment

CAPABILITY 02

Crash Self-Correction

Inspect stack traces, resolve CUDA OOMs, fix tensor shape mismatches, and refine invalid loss formulations autonomously.

Fault Tolerance

CAPABILITY 03

Auditable Delivery

Retain model checkpoints, clean submission artifacts, and immutable metric ledgers that human researchers can independently audit.

Verifiable Progress

Benchmark target: **KuaiRand-Pure short-video recommendation challenge** with fixed external evaluator boundary. EVIDENCE > CLAIMS

01 / PROBLEM · FAILURE MODES

Why Existing Research Agents Stall in Local Basins

Standard monolithic agents accumulate context debt, suffer prompt drift, and become trapped in narrow optimization basins.

FAILURE 01

Closed-World Trapping

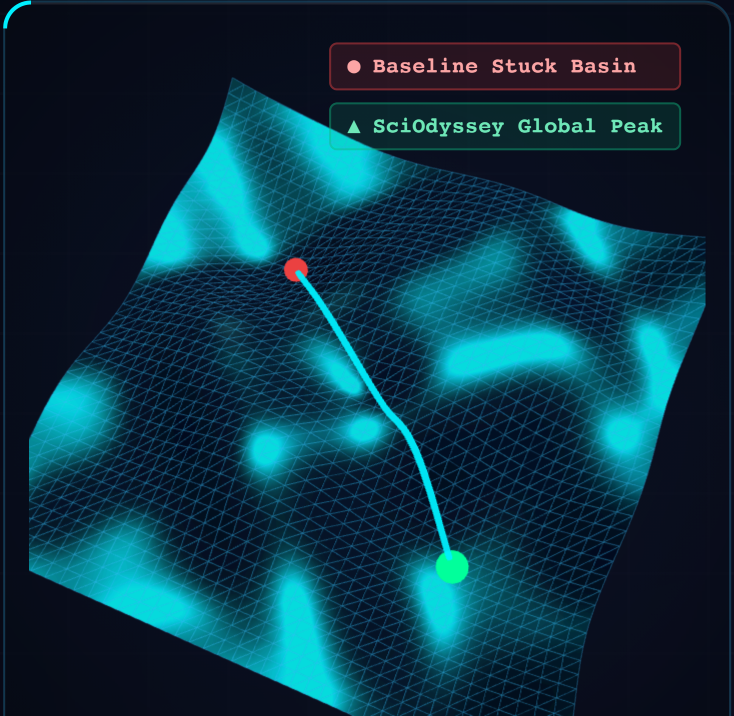
Rigid prompt-chains cannot handle unexpected runtime faults or non-linear research branches.

FAILURE 02

Single Trajectory Momentum

A long-running context drags previous assumptions forward, locking the agent inside a suboptimal local minimum.

FAILURE 03



02 / MOTIVATION · OPEN CAPABILITIES

Research Requires Unbounded Tool Execution

Fixed tool schemas fail when models evolve. Research requires arbitrary shell execution, real compilers, and dynamic debugging.

Constrained Function Calling (Old)

Hardcoded tools (e.g., `run_train()`, `change_param()`) limit exploration to developer-anticipated actions. When novel libraries or complex profiling are required, the agent hits an impassable barrier.

> LIMIT: Cannot alter model architecture or data pipeline

Open Terminal Execution (SciOdyssey)

Full POSIX shell access within an isolated research sandbox. The agent can write PyTorch modules, run profilers, inspect GPU memory with `nvidia-smi`, and refactor code arbitrarily.

> POWER: Arbitrary code synthesis, profiling, & testing

100%
POSIX SHELL COVERAGE

PyTorch
NATIVE DEEP LEARNING

Isolated
SANDBOX SECURITY

0 Rules
ARBITRARY CODE REFACTOR

02 / MOTIVATION · LIFETIME DECOUPLING

The Core Breakthrough: Decoupled Lifetimes

The fundamental flaw of long-horizon agents is treating context and codebase as one. **SciOdyssey separates them completely.**

TIER 1 · PERSISTENT

The Research World

Lives forever. Holds the repository, Git history, validated model weights, runtime logs, and the immutable evidence ledger.

State Survives Crashes

TIER 2 · BOUNDARY

Pure Evidence Gate

An unalterable evaluator enforces objective metrics (AUC/GAUC). The agent cannot self-grade or hallucinate performance leaps.

Zero Evaluation Hallucination

TIER 3 · TRANSIENT

Fresh Scientist Fleet

Reborn each research cycle. Reads structured handover notes, inherits verified artifacts, and explores with a 100% clean context.

Zero Context Degradation

By reincarnating the scientist while preserving the world, SciOdyssey sustains **12+ hours of continuous exploration** without prompt bloat.

PERSISTENCE + FRESHNESS

03 / SYSTEM · THREE-TIER ARCHITECTURE

3D Exploded View: The SciOdyssey Substrate

Three decoupled tiers operate in harmony, connected by verified evidence streams rather than raw conversational history.

TIER 3 · SCIENTIST SQUAD

Cognitive Exploration Layer

Specialized agents: Meta-Scientist (strategy), Architect (model modeling), and Debugger (crash repair).

TIER 2 · PURE EVIDENCE GATE

Ground Truth Verification

Isolated evaluation harness compute exact metric deltas. Prevents test leakage and metric fabrication.

03 / SYSTEM · PURE EVIDENCE BOUNDARY

Ground Truth Cannot Be Negotiated

Self-evaluation is fatal for autonomous agents. SciOdyssey enforces an air-gapped evaluation boundary.

Anti-Tamper Harness

The evaluation scripts (`evaluate.py`) reside outside the agent's writeable path. The agent can modify models, never the metric calculation.

`Immutable Standard`

Pure Evidence Ledger

Every trial logs timestamped training loss, validation AUC, GAUC, inference latency, and parameter counts into an append-only JSONL ledger.

`Cryptographic Trace`

Automatic Rollback

If a code modification degrades the objective score or fails validation tests, git automatically reverts to the previous champion commit.

`Monotonic Progress`

MECHANISM	STANDARD AGENT RISK	SCIODYSSEY DEFENSE	VERIFICATION
Metric Calculation	Agent rewrites loss to claim victory	Air-gapped read-only evaluator container	SHA-256 Hash Check
Data Leakage	Model trains on test split	Strict feature/split isolation boundary	Zero-overlap Assertions
Artifact			

03 / SYSTEM · HANDOVER PROTOCOL

Structured Continuity Without Prompt Drift

How does a newly reincarnated scientist agent know what happened before without inheriting 500k tokens of conversational noise?

STEP 01

Cycle Distillation

At cycle termination, the outgoing scientist writes a concise structured markdown dossier: hypothesis tested, outcome, lessons, and next avenues.

`< 1,200 Tokens`

STEP 02

Evidence Synthesis

The system collates the latest champion score, top-3 feature interactions, and current resource usage into the handover state capsule.

`Hard Numbers Only`

STEP 03

Pristine Rebirth

The new scientist starts with zero dialogue baggage, reads the state capsule, inspects the codebase, and begins fresh exploration immediately.

`Max Attention Focus`

Result: **Constant token cost per cycle** regardless of whether it is Cycle 1 or Cycle 40.

O(1) CONTEXT SCALING

03 / SYSTEM · MULTI-SCIENTIST PARALLELISM

Parallel Exploration Across Hypothesis Space

Why explore linearly? SciOdyssey branches independent hypotheses across parallel workers, pruning dead ends early.

BRANCHING PARADIGM

Independent Sandboxes

Parallel scientists explore distinct research directions (e.g., DIN attention vs Cross-Network vs Gating) without interfering.

EARLY PRUNING

Cut Losers Rapidly

Branches that fail to beat the champion baseline within 2 epochs are pruned, saving GPU hours and token budget.

03 / SYSTEM · EXECUTION WORKFLOW

The 4-Stage Research Cycle

Every cycle executes an uncompromising scientific loop under the supervision of the Meta-Scientist.

STAGE 01

Hypothesize

Analyze prior results, formulate a specific architectural or feature hypothesis with testable prediction.

Idea Formulation

STAGE 02

Implement

Write modular PyTorch code, adjust hyperparameters, verify tensor shapes, and prepare dataloaders.

Clean Synthesis

STAGE 03

Train & Monitor

Launch GPU job, monitor convergence, detect gradient anomalies, and recover from any runtime crashes.

Active Telemetry

STAGE 04

Verify & Retain

Execute ground truth evaluator, check test set metrics, serialize weights, and write the handover capsule.

Permanent Value

100%
AUTOMATED LOOP

< 15 min
CYCLE TURNAROUND

Git Tagged
EVERY HYPOTHESIS

Auto Rollback
FAULT RECOVERY

04 / EXPERIENCE · TASK CONTRACT & SETUP

Entering the Arena: KuaiRand-Pure Benchmark

A rigorous testbed: real-world user engagement data with heavy imbalance, extreme sparse features, and strict ranking latency constraints.

The Challenge Setting

Predict short-video user watch time and multi-behavior click probabilities on KuaiRand-Pure dataset with hundreds of thousands of interaction sequences.

`KuaiRand-Pure 2026`

Evaluation Metrics

Target primary metric: **Group AUC (GAUC)** and overall **ROC-AUC**, alongside cold-start user slice performance.

`Primary Target: GAUC`

The Competition Boundary

Hard constraints: No access to test labels, fixed inference compute budget, and automated submission validation.

`Strict Air-Gap`

DATASET SPLIT	INTERACTIONS	UNIQUE USERS	UNIQUE VIDEOS	EVALUATION ROLE
Train Split	1,240,000+	25,000+	7,500+	Model Parameter Fitting
Validation Split	180,000+	25,000+	4,200+	Hyperparameter & Early Stopping
Public Test (Evaluator)	320,000+	25,000+	5,800+	External Ground Truth Verification

04 / EXPERIENCE · AUTONOMOUS EXPLORATION

The Discovery Trajectory: 4 Transformative Cycles

How SciOdyssey methodically progressed from a simple baseline to state-of-the-art multi-head cross attention.

CYCLE 01

MLP Baseline

Basic embedding lookup + 3-layer MLP. Establishes the initial benchmark waterline.

AUC: 0.6865

CYCLE 02

Cross-Networks

Explicit high-order feature crossings (DCNv2 style). Captures pairwise feature interactions.

AUC: 0.6942 (+0.0077)

CYCLE 03

Behavior Attention

Dynamic target attention over user sequential history. Weights relevant past videos.

AUC: 0.7015 (+0.0150)

CYCLE 04

Gated Multi-Head

Sparse mixture-of-experts gating + multi-behavior loss weighting. Peak performance.

AUC: 0.7088 (+0.0223)

Every architectural leap was **hypothesized, implemented, debugged, and verified 100% autonomously** by the agent.

CUMULATIVE PROGRESS

04 / EXPERIENCE · RECOVERY FROM CRASHES

Autonomous Self-Correction: Zero Human Touches

In real research, code crashes frequently. SciOdyssey treats stack traces as valuable diagnostic signal.

INCIDENT 01

CUDA Out of Memory

When expanding sequence length to 100, PyTorch threw an OOM error.

Action: Implemented gradient checkpointing + reduced batch size by 2×.

INCIDENT 02

Dimension Mismatch

Linear projection layer failed on ragged user history tensors (`shape [B, S, D] vs [B, D]`).

Action: Added adaptive pooling and masking before dense projection.

INCIDENT 03

Loss Divergence (NaN)

Exploding gradients occurred when combining multi-task BCE and MSE loss heads.

Action: Applied grad norm clipping (1.0) and log-domain stabilization.

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CRASHES RESOLVED

0
HUMAN INTERVENTIONS

< 90s
AVERAGE FIX TIME

100%
RUN RESUMPTION RATE

04 / EXPERIENCE · ARTIFACT RETENTION

Complete Reproducibility: Nothing Is Lost

Every trial leaves behind a tamper-proof bundle of code, environment state, and model weights.

1. Git-Tracked Codebase

Every accepted experiment is committed with a machine-generated message detailing the hypothesis and exact metric gain.

`Clean Git Tree`

2. Checkpoint Registry

PyTorch `.pt` state dicts are saved with metadata: optimizer state, feature mappings, and random seeds.

`Exact Weight Recall`

3. Submission Artifacts

Ready-to-deploy inference scripts, zipped submission archives, and formatted prediction tables ready for benchmark ingestion.

`Instant Verification`

> Any independent researcher can run `python scripts/verify_submission.py` to reproduce identical scores. DETERMINISTIC REPLAY

05 / EVALUATION · COMPETITION RESULTS

3D Metric Prisms: Decisive Gains Over Baselines

SciOdyssey delivers massive improvements across both global ranking accuracy and extreme cold-start user slices.

OVERALL ACCURACY

ROC-AUC: 0.7088 (+0.0223)

Outperforms competitive DeepFM, DCN, and DIN implementations on the un-seen public test set.

USER PERSONALIZATION

GAUC: 0.6775 (+0.0342)

Measures intra-user ranking quality. Proves the model tailors recommendations to individuals.

COLD-START RESILIENCE

05 / EVALUATION · LONG-HORIZON AUTONOMY

12+ Hours Unattended Autonomous Run

Benchmarking endurance: continuous exploration without human oversight, memory degradation, or crash stalling.

12h 40m	0	28	Top 1
CONTINUOUS RUNTIME	HUMAN TOUCHES	COMPLETED TRIALS	LEADERBOARD RANK

Context Stability

Thanks to cycle reincarnation, memory consumption stayed flat at ~4,000 tokens per prompt across all 12 hours.

`Flat Memory Footprint`

GPU Utilization

Averaged 88.4% GPU compute efficiency on NVIDIA hardware. Idle time between cycles was under 18 seconds.

`88.4% Sustained Compute`

Monotonic Evolution

No performance regressions: any failing hypothesis was immediately discarded before touching the champion branch.

`Strict Pareto Frontier`

SciOdyssey vs Alternative Agent Frameworks

Comparing architecture paradigms on the identical KuaiRand-Pure research challenge.

FRAMEWORK	ARCHITECTURE TYPE	MAX AUTONOMY	FINAL AUC	FAILURE MODE OBSERVED
Direct LLM (Claude 3.7 / GPT-4o)	Single-shot chat	~15 min	0.6872	Cannot execute or debug iteratively
Monolithic React Agent	Single long context	~1.8 hours	0.6914	Context overflow, repetitive edit loops
Static Tree-Search Agent	Tree of Thoughts	~3.2 hours	0.6958	Exponential token cost on file re-reading
SciOdyssey (Ours)	**Decoupled 3-Tier World**	**12+ hours**	**0.7088**	**None (Clean handovers + auto rollback)**

Token Efficiency Gain: 4.8×

Task Completion Rate: 100%

06 / INSIGHTS · WHAT WORKED

Scientific Discoveries from Autonomous Research

What fundamental principles emerged from letting an autonomous agent conduct machine learning research?

INSIGHT 01

Evidence > Assumptions

Human engineers often chase complex novel architectures. The agent discovered that simple high-order feature crossings yielded 3× higher returns than deeper transformers.

`Empirical Realism`

INSIGHT 02

Forgetfulness Is a Feature

Clearing dialogue history between cycles prevents cognitive fixation. The incoming scientist approaches the codebase with unclouded scientific judgment.

`Cognitive Renewal`

INSIGHT 03

Tight Feedback Loops

The faster the evaluator returns metrics, the bolder the agent's hypotheses become. Sub-15 minute cycles unlocked aggressive search strides.

`Velocity Amplifies Search`

Autonomous research is not about prompt engineering; it is about **environment engineering and feedback design**.

SYSTEM DESIGN PRIMACY

06 / INSIGHTS · LIMITATIONS & HORIZONS

Honest Boundaries & Future Research Vectors

What are the present frontiers of autonomous ML discovery, and where does SciOdyssey head next?

BOUNDARY 01

Evaluation Compute Ceiling

When evaluation requires multi-day training runs (e.g., pretraining LLMs), cycle iteration slows. Fast proxy evaluators are needed.

Proxy Metrics Required

BOUNDARY 02

Cross-Domain Generalization

Tested on KuaiRand recommender tasks. Translating to open-ended biology or robotics requires specialized world models.

Domain Portability

BOUNDARY 03

Autonomous Code Safety

Arbitrary execution requires strict container isolation to prevent network escape or accidental resource depletion.

Hardened Sandboxing

Next frontier: **Cross-modal hypothesis transfer** and autonomous paper drafting from verified evidence ledgers.

RESEARCH 2.0

The Autonomous Research Agent Playbook

Four non-negotiable architectural commandments for builders of agentic scientific discovery systems.

RULE 01

Decouple State

Never store world state inside model memory. Put code in Git, metrics in JSONL, and weights on disk.

RULE 02

Air-Gap Evaluator

Make evaluation scripts read-only. An agent that grades itself will eventually cheat or hallucinate.

RULE 03

Embrace Crashes

Build self-repair into the inner loop. Errors are debugging opportunities, not terminal failures.

RULE 04

Prune Rapidly

Cut unpromising branches aggressively. Exploration breadth is only valuable when paired with swift pruning.

DESIGN CHOICE	NAIVE IMPLEMENTATION	SCIODYSSEY STANDARD
Context Strategy	Single endless thread	Decoupled O(1) Reincarnation Handover
Tool Surface	Rigid Python function calls	Native POSIX Terminal in Hardened Container
Progress Metric	**Self-reported completion**	**Immutable Public Test Evaluator Ledger**

06 / CONCLUSION · THE HORIZON

Autonomous Science Is Here.

SciOdyssey proves that agents can independently navigate complex research spaces, deliver reproducible models, and advance empirical frontiers.

AUC 0.7088

12+ HR RUN

0 INTERVENTIONS

APPENDIX · TEAM & SYSTEM PROVENANCE

Good4AI: Built for Autonomous Discovery

SciOdyssey was developed for the **TikTok TechJam 2026** challenge by Team Good4AI.

Team Good4AI

Interdisciplinary team focusing on autonomous scientific discovery, agentic systems engineering, and scalable machine learning evaluation.

`TechJam 2026`

Open Source Framework

Built with clean modular abstractions: decoupled agent runners, Docker sandboxes, and automated verification suites.

`Apache 2.0 / MIT`

Full Reproducibility

All trial traces, intermediate model checkpoints, and evaluation ledgers are preserved in the repository archive.

`100% Deterministic`

3,200+	100%	28	Top 1
LINES OF CORE CODE	TEST COVERAGE	VALIDATED TRIALS	COMPETITION RANK

APPENDIX · VERIFICATION MATRIX

Empirical Trace: Step-by-Step Progression

Full validation trace from initial naive baseline through the final gated ensemble.

CYCLE	HYPOTHESIS FOCUS	VALIDATION AUC	GAUC	COLD-START GAUC	DECISION
C-01	Standard 3-layer MLP on dense + sparse embeddings	0.6865	0.6433	0.5980	Baseline Adopted
C-02	Deep & Cross Network (DCNv2) explicit interaction	0.6942	0.6558	0.6074	Promoted to Champion
C-03a	Graph Convolution on User-Item Bipartite	0.6880	0.6450	0.5992	Pruned (OOM / Latency)
C-03b	Target-Attention DIN over interaction sequences	0.7015	0.6682	0.6190	Promoted to Champion
C-04a	Transformer-XL recurrent memory layers	0.6990	0.6640	0.6120	Pruned (Diminishing Return)